



Interuniversity Program in Water Resources Engineering

MSc Thesis Proposal

Evaluating Satellite-Based Rainfall Estimates for Flood Modelling and Early Warning in the Ky
Lo River Basin, Vietnam

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I. INTRODUCTION

Water-Related Disasters (WRD) account for an overwhelming 90% of all natural disasters globally. Among these, floods are the most frequent, representing about 54% of all WRDs and causing massive economic damage and loss of life. Asia is currently the most impacted continent, accounting for 41% of all global flood disaster events. Consequently, the implementation of Flood Early Warning Systems (FEWS) has been strongly advocated by the Sendai Framework for Disaster Risk Reduction and the Sustainable Development Goals (SDGs) as a critical non-structural measure to mitigate flood impacts. According to the United Nations Office for Disaster Risk Reduction (UNDRR), an effective FEWS requires four key elements: risk knowledge, monitoring and forecasting, warning dissemination and response capabilities. However, many developing countries face substantial challenges in the forecasting stage due to technical and infrastructural limitations. A global survey of 53 countries found that while the frequency of operational FEWS has doubled since 2000, approximately 75% of river basins globally still report inadequate rainfall and water-level gauging networks within their basins, which restricts their ability to generate reliable flood forecasts. (Perera et al., 2019)

The Ky Lo River basin along Vietnam's South-Central Coast serves as a primary example of this vulnerability. The basin spans 2,058 km² with a main river length of 103 km and an average width of 15.8 km. Its hydrological behavior is strongly influenced by the Cu Mong and Deo Ca mountain ranges, which form a valley where terrain height gradually decreases from west to east. This topographic gradient promotes concentrated runoff and increases the occurrence of flash floods. Historical records indicate that more than 85% of flood events occur during October and November. Climate change projections under the RCP 8.5 scenario further suggest that the maximum flooded area in the downstream Ky Lo Basin could expand to approximately 9,562 hectares by 2070. (Bach Tung et al., 2021) These factors underscore the urgent need for advanced monitoring and forecasting capabilities to improve flood preparedness and safeguard vulnerable communities in the region.

Despite the clear and increasing flood risk, the Ky Lo River basin remains a data-scarce region. While several automatic rain gauge stations have recently been installed within the basin, these are primarily deployed in accessible low-lying areas. In high mountainous areas where infrastructure is limited, rain gauge equipment has not yet been deployed. Crucially, heavy rainfall is often concentrated in these upland regions, resulting in a severe lack of ground-based rainfall monitoring and early warning information where it is most needed. This spatial gap makes real-time flood warning exceptionally challenging as the existing gauge network fails to capture the high spatial and temporal variability of precipitation caused by the basin's complex topography.

To bridge this observational gap and address the lack of ground-based observational data for operational FEWS, Satellite-derived Precipitation Estimates (SPEs) have emerged as a promising alternative due to their high-resolution and near-real-time rainfall monitoring capabilities. Products such as the Global Precipitation Measurement (GPM), Climate Hazards

Group InfraRed Precipitation (CHIRP) and Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks (PERSIANN) provide extensive spatial and temporal rainfall coverage, making them particularly valuable for ungauged or data-scarce regions. (Le et al., 2020)

Previous research across multiple Vietnamese basins has demonstrated that GPM IMERG Final Run (IMERGF-V6) achieved the best overall performance among SPE products for both rainfall detection and streamflow simulation, while CHIRPS demonstrated the lowest bias, favoring long-term water resources planning (Le et al., 2020). The same study found that in basins with low rain gauge density, SPE-driven hydrological simulations using the Soil and Water Assessment Tool (SWAT) outperformed gauge-driven simulations, highlighting the potential of satellite data where ground networks are inadequate (Le et al., 2020). In Northern Vietnam, CHIRPS and PERSIANN-CDR also demonstrated utility for long-term trend assessments (Nguyen et al., 2025).

However, despite their accessibility and increasing application in hydrological studies, SPEs remain under-utilized in local flood forecasting operations (Perera et al., 2019). This is primarily because satellite sensors estimate rainfall indirectly, introducing uncertainties associated with retrieval algorithms, climatic conditions, and complex topographic features. As a result, systematic evaluation, bias correction and local validation against ground-based observations are essential before SPEs can be reliably applied in operational flood forecasting. Previous studies have shown that bias-corrected SPEs can significantly improve hydrological simulations, particularly when integrated with physically based models such as the Soil and Water Assessment Tool (SWAT), which has demonstrated strong performance in hydrologic assessments throughout Vietnam. (Belay et al., 2019)

This research aims to evaluate the applicability of early-run or near-real-time SPE products for flood forecasting in the Ky Lo River Basin. By applying bias-correction techniques using available ground rainfall stations, the study seeks to determine whether SPEs can effectively drive hydrological models for flood mitigation and early warning applications in data-scarce Vietnamese river basins.

II. RESEARCH QUESTIONS

1. Which near-real-time SPE product (IMERG Early Run, CHIRP, or PERSIANN) most accurately represents rainfall patterns in the data-scarce Ky Lo River Basin when validated against available ground-based observations?
2. To what extent can bias correction improve the performance of satellite rainfall products for hydrological simulation in this basin?
3. How accurately do SWAT and HEC-HMS capture extreme flood peaks and streamflow dynamics when driven by bias-corrected SPEs, and how useful are these models for early warning applications?
4. Is there sufficient lead time from near-real-time satellite precipitation products to support an operational flood early warning system in the Ky Lo Basin?

III. OBJECTIVES

1. To extract, harmonize, and compare multi-source satellite rainfall products (IMERG Early Run, CHIRP and PERSIANN) using Google Earth Engine (GEE) and Geographic Information System (GIS).
2. To validate the performance of these SPE datasets against ground-based rain gauge observations (2021-Present) and apply bias correction techniques to improve the accuracy of the satellite rainfall data prior to model integration.
3. To conduct a comparative hydrological simulation using SWAT and HEC-HMS, evaluate the accuracy of each model's flood peak and streamflow simulations, and assess their suitability for early flood warning.
4. To evaluate the operational utility of near-real-time satellite products for Early Warning Systems (EWS), specifically analyzing the trade-off between satellite expectancy of 1 to 4 hours and flood travel time of 6 to 10 hours

IV. METHODOLOGY

The proposed research will employ a multi-stage technical framework to evaluate the integration of near-real-time Satellite-derived Precipitation Estimates (SPE) into a Flood Early Warning System (FEWS). The workflow is divided into data harmonization, statistical bias correction, comparative hydrological modeling and operational lead-time evaluation.

Study Area

Based on the availability of essential hydrological data, the Ky Lo River Basin is selected as the primary study area. This basin is particularly suitable because it possesses both consistent rainfall data from 2021 to present and discharge data from 2019 to 2023. These are critical for accurate hydrological calibration and validation. The Ky Lo Basin is characterized by steep topography and a humid tropical monsoon climate, making it highly susceptible to severe flooding. (Phan et al., 2025)

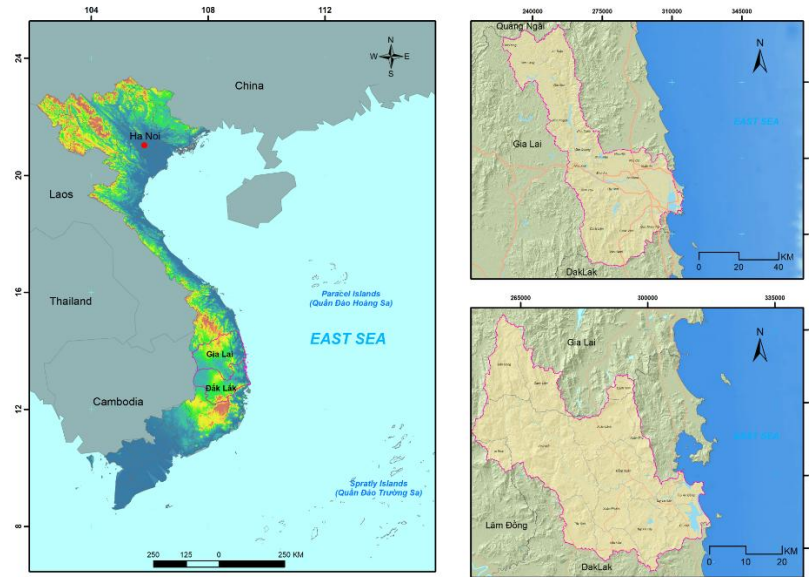


Figure 1. Study area.

The Ky Lo River basin lies between latitude $13^{\circ}09'15''$ - $13^{\circ}46'40''$ North and longitude $108^{\circ}42'08''$ - $109^{\circ}19'08''$ East, stretching across Dong Xuan and Tuy An districts of the former Phu Yen province, now part of Dak Lak province, with a portion extending into the former Binh Dinh and Kon Tum provinces, now part of Gia Lai and Quang Ngai provinces. The basin area up to the estuary is 2,058 km² with a main river length of 103 km and an average basin width of 15.8 km. The basin is part of the South-Central Coastal region, characterized by a humid tropical monsoon climate with two distinct seasons: a dry season from January to August and a rainy season from September to December. Annual rainfall during the rainy season ranges from 1,152 to 1,738 mm, accounting for 68–84% of total annual precipitation while the dry season receives 260-684 mm. The basin's topography is primarily governed by the Cu Mong and Deo Ca mountain ranges and the Ky Lo River valley, with terrain gradually decreasing from west to east, promoting concentrated runoff and flash flooding. River discharge mirrors rainfall distribution, with the four flood months accounting for 70-75% of annual discharge, while the eight dry months account for approximately 25-30% (Bach Tung et al., 2021).

The Ky Lo Basin is particularly suitable for this study because it possesses consistent rainfall data from 2021 to present and discharge data from 2019 to 2023. These data are critical for accurate hydrological calibration and validation. Previous studies have successfully applied hydrodynamic modeling namely, MIKE FLOOD, to simulate flooding in this basin with calibration and validation against major flood events in November 2009 and November 2010 (Bach Tung et al., 2021). Additionally, recent flood mapping studies using Sentinel-1 SAR imagery and Google Earth Engine have demonstrated the effectiveness of remote sensing approaches for flood extent delineation in the Ky Lo basin (Phan et al., 2025). These studies provide baseline data and methodological precedents for current research.

Data Acquisition & Pre-processing

The study will utilize Google Earth Engine (GEE) and Python-based automation to extract Satellite-derived Precipitation Estimates (SPEs). This approach is consistent with recent flood mapping studies in the Ky Lo basin, which highlight GEE's efficiency in handling large-scale remote sensing datasets (Phan et al., 2025). The study will focus on three SPE products identified as having strong potential for driving hydrological models in Vietnam's data-scarce regions (Le et al., 2020) (Thom et al., 2017):

1. GPM IMERG Early Run V06

The IMERG Early Run product is characterized by a spatial resolution of 0.1° and a high half-hourly temporal frequency. With a latency of approximately four hours, this version provides near-real-time data that is particularly well-suited for operational forecasting. While previous evaluations in Vietnam indicated that the IMERG Final Run yields the highest overall accuracy, the Early Run remains the most practical configuration for real-time applications (Le et al., 2020).

2. CHIRP V2.0

CHIRP offers a finer spatial resolution of 0.05° on a daily temporal scale. Although it possesses a latency of several days, its high level of detail and characteristically low bias make it an ideal candidate for long-term environmental assessments and as a benchmark for comparative studies (Alejo & Alejandro, 2021; Le et al., 2020).

3. PERSIANN (0.25° , hourly, several-day latency)

The PERSIANN system provides data at an hourly temporal resolution with a spatial grid of 0.25° . Like CHIRP, it operates with a latency of several days. Previous studies within the Vietnamese context have yielded mixed performance results for PERSIANN, highlighting the necessity for local validation before its integration into broader hydrological models (Le et al., 2020; Thom et al., 2017).

The methodology includes standardization of the Digital Elevation Model (DEM), land use/land cover (LULC), and soil data using consistent spatial resolutions. Previous studies in the Ky Lo basin highlight the importance of these parameters for simulating two-dimensional flood flow in horizontal directions across inundated areas (Bach Tung et al., 2021; Phan et al., 2025)

Statistical Performance Assessment

The reliability of each SPE product will be evaluated against ground-based gauge observations across two complementary tiers of metrics, following approaches established for SPE validation in Vietnam and comparable data-scarce tropical (Muñoz De La Torre et al., 2024; Nguyen et al., 2025). This dual-framework design ensures that satellite products are assessed not only on how accurately they represent rainfall, but also on whether that accuracy translates into reliable hydrological simulations.

Accuracy Metrics for Rainfall Estimates

Statistical comparison between SPE-derived and gauge-observed rainfall will be conducted using three indicators. The Root Mean Square Error (RMSE) quantifies the overall magnitude of deviation between satellite estimates and gauge observations, with greater sensitivity to large errors making it particularly relevant for detecting failures during high-intensity events. The Pearson Correlation Coefficient (CC) measures temporal agreement between satellite and gauge time series, capturing how well each product reproduces the timing and variability of rainfall rather than just its volume. The Relative Bias (RB) isolates systematic directional error, persistent overestimation or underestimation which propagates directly into hydrological model inputs and degrades flood simulation reliability if left uncorrected. Evaluated together, these three metrics characterize the magnitude, temporal structure, and systematic bias of each SPE product prior to bias correction.

Hydrological Performance Metrics

Following bias correction and model integration, simulated streamflow will be evaluated against observed discharge records using four widely applied indicators. The Nash-Sutcliffe Efficiency (NSE) assesses the overall predictive skill of the model relative to using the observed mean as a benchmark with sensitivity to peak flow errors. The Kling Gupta Efficiency (KGE) assigns the model performance into correlation, variability ratio, and bias ratio, providing a more balanced assessment than NSE alone. The Percent Bias (PBIAS) quantifies volumetric over or under prediction across the full simulation period. The fourth indicator, the RMSE-Observations Standard Deviation Ratio (RSR) normalizes RMSE by the standard deviation of the observed flow record, enabling performance comparisons across sub-catchments and flow regimes with differing hydrological variability.

Bias Correction

Prior to hydrological model integration, systematic biases identified in the SPE products will be corrected using available ground rainfall station data. Bias correction methods will follow approaches demonstrated in previous studies to significantly improve hydrological simulations in data-scarce regions (Belay et al., 2019)

Hydrological Model Implementation

Both SWAT and HEC-HMS models will be set up and calibrated using the period from 2021 to late 2023. The SWAT model has been widely validated in Vietnam, including applications in the Srepok River Catchment where it achieved satisfactory streamflow simulations using satellite rainfall inputs. Calibration will be performed using the SUFI-2 algorithm within SWAT-CUP, following established protocols. (Thom et al., 2017)

Given that HEC-HMS is often more robust for extreme event simulation, it will be compared against SWAT to evaluate the accuracy of each model's flood simulations and determine their

suitability for flood forecasting. The model's performance will be further tested using historical flood event information to evaluate its accuracy under real flood conditions. To verify forecasting potential, the model will be run to predict flow for December 2023, comparing results directly against available gauge data to assess short-term reliability.

Early Warning Framework Evaluation

The final stage evaluates the feasibility of a satellite-based Early Warning System (EWS) for the basin. The study factors in the basin's 6 to 10 hour flood travel time, estimated based on the river's 110 km length and steep terrain, against the latency of SPE products. GPM IMERG Early Run provides near-real-time data with approximately 4 hours of latency, potentially offering a critical 2-to-6-hour lead-time window for warning dissemination and response. The research will conclude by recommending rainfall thresholds for local EWS to improve resilience against future flood-related losses.

V. TIMELINE

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VI. REFERENCES

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